

Regression Assignment

Problem Statement :

predict the insurance charges based on the several parameters.

DataSet:

https://raw.githubusercontent.com/RamishaRaniK/dataset/main/insurance_pre.csv

Rows: 1338, Columns: 6

Use One-Hot Encode Categorical (Sex,Smoker) columns.

Machine Learning Regression method using R^2 value :

1. Multiple Linear Regression (R^2 Value) = 0.7894
2. Support Vector Machine:

S.No	Hyper Parameter	Linear (r value)	RBF (Non Linear) (r value)	POLY (r value)	SIGMOID (r value)
1	C=0.1	-0.12207	-0.089576	-0.0862525	-0.08974
2	C=10,	-0.0016176	-0.08196	-0.09311	-0.0907
3	C=100	0.54328	-0.124803	-0.09976	-0.1181
4	C=1000	0.63403	-0.117490	-0.055505	-1.6659
5	C=1000, degree=3, epsilon=0.1, coef0=1	0.634036	-0.1174	0.0096478	-0.153963
6	C=10000	0.74448		0.3529	-119.51859

With Standard Scaler:

S.No	Hyper Parameter	Linear (r value)	RBF (Non Linear) (r value)	POLY (r value)	SIGMOID (r value)
1	C=0.1	-0.08095	-0.08907	-0.0883	-0.08826
2	C=10,	0.46246	-0.03227	0.03871	0.03930
3	C=100	0.62887	0.3200	0.617956	0.5276
4	C=1000	0.7649	0.8102	0.85664	0.2874
5	C=1000, degree=3, epsilon=0.1, coef0=1	0.764931	0.8102	0.873519	-1.3731

3. Decision Tree:

criterion	splitter	max_depth	min_samples_split	min_samples_leaf	r2_score
absolute_error	best	5	5	2	0.886096
absolute_error	best	5	5	4	0.886096
absolute_error	best	5	10	4	0.886096

absolute_error	best	5	2	2	0.886096
absolute_error	best	5	2	4	0.886096
absolute_error	best	5	10	2	0.886096
absolute_error	best	10	10	4	0.884566
absolute_error	best	20	10	4	0.884476

4. Random Forest

n_estimators	max_depth	min_samples_split	min_samples_leaf	max_features	bootstrap	r2_score
200	5	5	4	None	TRUE	0.891514
200	5	2	4	None	TRUE	0.891514
200	5	10	4	None	TRUE	0.891509
100	5	2	4	None	TRUE	0.891282
100	5	5	4	None	TRUE	0.891282
50	5	2	4	None	TRUE	0.891248
50	5	5	4	None	TRUE	0.891248
100	5	10	4	None	TRUE	0.891241
50	5	10	4	None	TRUE	0.891201
200	5	10	2	None	TRUE	0.890522

5. XGBoost:

n_estimators	learning_rate	max_depth	min_samples_split	min_samples_leaf	subsample	loss	r2_score
100	0.05	3	10	1	0.7	huber	0.89389
100	0.05	3	2	1	0.7	huber	0.893865
100	0.05	3	5	1	0.7	huber	0.893863
300	0.01	5	10	4	0.7	huber	0.893787
100	0.05	5	10	4	0.7	huber	0.893737
300	0.01	5	2	4	0.7	huber	0.893651
300	0.01	5	5	4	0.7	huber	0.893651

100	0.05	3	5	4	0.7	hub er	0.8936 35
100	0.05	3	2	4	0.7	hub er	0.8936 35
100	0.05	3	10	4	0.7	hub er	0.8936 17

Conclusion:

The best-performing model in the entire experiment was

GradientBoostingRegressor, achieving an impressive R^2 score of 0.893.